HAOLIN YANG

hy3217@nyu.edu +1 7732567911 ◇ Personal Website

EDUCATION

New York University **Sept. 2026 – Present**
Ph.D. in Data Science

University of Chicago **Sept. 2024 – June 2026**
Master of Science in Statistics (GPA: 3.96/4.0)

Tsinghua University **Sept. 2020 – June 2024**
Bachelor of Arts in English (GPA: 3.94/4.0)
Minor in Economics and Finance (GPA: 4.0/4.0)

PUBLICATIONS AND MANUSCRIPTS

Preprints

1. Yan, H., **Yang, H.**, & Zhong, Y. (2026). Task Vector Geometry Underlies Dual Modes of Task Inference in Transformers. Submitted to **NeurIPS 2026**.

2. Cho, H., **Yang, H.**, Kurkoski, B. M., & Inoue, N. (2025). Binary Autoencoder for Mechanistic Interpretability of Large Language Models. Submitted to **ICML 2026**. [Link]

Conferences

1. **Yang, H.**, Cho, H., Zhong, Y., & Inoue, N. (2025). Unifying Attention Heads and Task Vectors via Hidden State Geometry in In-Context Learning. In **NeurIPS 2025**. [Link]

2. **Yang, H.**, Cho, H., & Inoue, N. (2025). Localizing Task Recognition and Task Learning in In-Context Learning via Attention Head Analysis. In **ICLR 2026**. [Link]

3. **Yang, H.**, Cho, H., Ding, K., & Inoue, N. (2025). Task Vectors, Learned Not Extracted: Performance Gains and Mechanistic Insights. In **ICLR 2026**. [Link]

4. Cho, H., **Yang, H.**, Minegishi, G., & Inoue, N. (2025). Mechanism of Task-Oriented Information Removal in In-Context Learning. In **ICLR 2026**. [Link]

5. Zhao, H., Zhou, S., **Yang, H.**, Qin, Z., & Zhou, T. (2026). Neuro-Symbolic Synergy for World Modeling. In **COLM 2026**. [Link]

Journals

1. Zhang, Y., Yang, F., **Yang, H.**, & Han, S. (2024). Does Checking-In Help? Understanding L2 Learners' Autonomous Check-In Behavior in an English-Language MOOC through Learning Analytics. **ReCALL**, 36(3), 343–358. [Link]

SERVICE

Reviewer, International Conference on Learning Representations **(ICLR 2025)**

HONORS AND AWARDS

Academic Excellence Scholarship, Tsinghua University (2022–2023)
Outstanding Undergraduate, Tsinghua University (2024)
Outstanding Undergraduate Thesis, School of Humanities, Tsinghua University (2024)

SKILLS AND LANGUAGES

Programming: Python (PyTorch, Transformers), R, Stata
Languages: Mandarin (native), English (fluent), German (C1)